



(10) Publication No: **IN202641068105 A1**

(22) Date of Filing: 30-05-2026

(43) Publication Date: 05-06-2026

Journal No: 23/2026

(19) Intellectual Property Office, India

(12) INDIAN PATENT APPLICATION

(54) Title: **System for risk-based stateful control of artificial intelligence agent interactions**

(51) International Classification:

**G06N 20/00, G06N 5/04, G06N 5/02, G06N 3/08
G06N 3/04**

(21) Application No: **202641068105**

(31) Priority Document No: --

(32) Priority Date: --

(86) International Application No: --

Filing Date: --

(87) International Publication No: --

(61) Patent of Addition to Application No: --

Filing Date: --

(62) Divisional to Application No: --

Filing Date: --

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(57) Abstract:

ABSTRACT SYSTEM FOR RISK-BASED STATEFUL CONTROL OF ARTIFICIAL INTELLIGENCE AGENT INTERACTIONS A system for risk-based stateful control of artificial intelligence agent interactions comprises a processor (100) and memory (102) storing instructions causing the processor (100) to receive user prompts and interaction events via an input processing module (104), perform semantic analysis and context extraction to identify input intent and structure, compute via a risk computation engine (106) a continuous risk score using a weighted multi-parameter function incorporating semantic deviation scores, instruction consistency gradients, context sensitivity coefficients, and temporal interaction weights, determine via a state machine controller (108) a current security state from safe, suspicious, under attack, and lockdown states based on the computed risk score and temporal interaction patterns, regulate via a context exposure module (110) access to internal system prompts and memory representations, and execute via a multi-layer enforcement module (112) a control action comprising allow, sanitize, restrict, or block. (Fig. 1)

FORM-2

**THE PATENTS ACT, 1970
(39 of 1970)**

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THE PATENTS RULES, 2003

**COMPLETE
SPECIFICATION**

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(See section 10 and rule 13)

TITLE OF INVENTION

15 **SYSTEM FOR RISK-BASED STATEFUL CONTROL OF ARTIFICIAL
INTELLIGENCE AGENT INTERACTIONS**

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25 **THE FOLLOWING SPECIFICATION PARTICULARLY DESCRIBES
THE INVENTION AND THE MANNER IN WHICH IT IS TO BE
PERFORMED**

CROSS-REFERENCE TO RELATED APPLICATIONS AND PRIORITY

[0001] The present application does not claim priority from any patent application.

PREAMBLE

5 [0002] The following specification particularly describes the invention and the manner in which it is to be performed.

FIELD OF INVENTION

[0003] The present disclosure relates to security systems for artificial intelligence agents, and more particularly to a system for risk-based stateful control
10 of artificial intelligence agent interactions using continuous risk scoring, state machine-based security management, and multi-layer enforcement mechanisms to protect against adversarial manipulation.

BACKGROUND

[0004] The following text of background art is provided for purposes of understanding and is not an admission that any cited document or technique forms
15 part of the prior art under applicable law.

[0005] Large Language Model (LLM)-based artificial intelligence agents have become increasingly prevalent in interactive applications, including customer service systems, autonomous agent platforms, and conversational interfaces. As
20 these AI agents are deployed in security-sensitive environments, they face growing threats from adversarial manipulation techniques such as prompt injection, instruction override attacks, and context leakage attempts.

[0006] Existing security mechanisms for LLM-based AI agents predominantly rely on static rule-based filtering, prompt sanitization techniques, or isolated
25 machine learning classifiers operating on individual input-output instances. While such approaches may mitigate explicit prompt injection attempts, they exhibit limitations when applied to adaptive, multi-step, and context-dependent adversarial interactions.

[0007] CN-121561061-A discloses a multi-round dialogue security defense
30 method that obtains multi-round dialogue content and tool invocation sequences, constructs a dialogue semantic graph representing semantic evolution and intent

flow, and performs collaborative risk judgment using preset rules to determine defense results. However, this approach relies on graph neural network-derived feature vectors with binary classification output and does not implement explicit computational state representations with defined transition conditions.

5 **[0008]** CN-121262016-B discloses a defense method for LLM prompt attacks that receives user input, generates multiple detection input copies, and inputs them in parallel into independent detection processes to obtain multiple risk scores. The system constructs a comprehensive risk score through weighted fusion and executes active defense strategies including instruction obfuscation and semantic dilution.
10 However, this approach uses stateless risk level intervals evaluated per-input and does not maintain persistent security states across interactions.

[0009] CN-121561909-A discloses an LLM attack defense method that performs malicious instruction risk assessment and consistency risk assessment, then performs weighted linear fusion to determine risk level and execute graded
15 defense strategies. CN-121765054-A discloses a multi-round dialogue risk detection method that generates dialogue snapshots, performs entity extraction and resolution, and determines dialogue risk level based on instruction type and content type using a preset risk level lookup table.

[0010] These existing solutions treat security enforcement as a post-processing
20 or filtering task rather than as a continuous control problem governing AI agent behavior across the entire interaction lifecycle. When adversarial interactions are not captured and systematically analyzed through stateful control mechanisms, AI agents may be susceptible to staged manipulation patterns, context leakage, and instruction override attacks. Accordingly, there remains a desire for improved
25 systems and methods for securing AI agent interactions.

[0011] The present invention addresses the above shortcomings of the prior art. However, the invention is entirely different from the prior art in terms of novelty and technological advancements.

OBJECT

30 **[0012]** In view of the foregoing, and without prejudice to the generality of the disclosure, it is an object of the present invention to provide. It is to be understood

that the foregoing objects are illustrative and non-limiting; additional objects will be apparent from the description, examples, and claims, and need not be achieved in every embodiment.

5 **[0013]** The object of the present disclosure is to provide a system for risk-based stateful control of artificial intelligence agent interactions that overcomes the limitations of existing security mechanisms. This system aims to implement continuous risk scoring combined with state machine-based security management to enable persistent tracking of security conditions across multiple interaction instances, rather than treating each input in isolation.

10 **[0014]** A further object of the present disclosure is to provide a multi-layer enforcement architecture that coordinates security actions across input filtering, reasoning control, tool execution regulation, and output validation layers under a unified control policy. By governing these layers through a common policy derived from computed risk scores and current security states, the system seeks to provide
15 comprehensive protection against adversarial manipulation techniques including prompt injection, instruction override attacks, and context leakage attempts.

[0015] Another object of the present disclosure is to provide a context exposure regulation mechanism that selectively controls access to internal system prompts and memory representations based on dynamically computed risk assessments. This
20 approach aims to prevent unauthorized disclosure of sensitive system information during adversarial interactions while maintaining normal operational functionality during benign interactions.

[0016] Yet another object of the present disclosure is to provide an attack memory and feedback adaptation mechanism that stores historical adversarial
25 interaction patterns and detected attack signatures, and utilizes this stored information to enhance risk prediction and enable proactive identification of multi-turn attack strategies. Through feedback-based parameter adjustment, the system aims to adapt its risk computation over time based on monitored outputs and interaction outcomes.

30 **[0017]** For the avoidance of doubt, the foregoing objects encompass compositions and manufacturing processes suitable for adoption in jurisdictions

where claims to methods of treatment of animals may be restricted; corresponding “use” or “product” claim formats are contemplated without limitation.

SUMMARY

[0018] The following summary is intended to introduce aspects of the present invention in a simplified form and does not identify all the features or the scope of the claimed invention. The summary is provided to assist in understanding the invention and is to be read in conjunction with the detailed description and accompanying drawings, claims, and abstract. One or more embodiments of the present disclosure are set forth below solely for illustrative purposes and should not be construed as limiting.

[0019] According to an aspect of the present disclosure, a system for risk-based stateful control of artificial intelligence agent interactions is provided. The system comprises a processor (100) and a memory (102) coupled to the processor (100). The memory (102) stores instructions that, when executed by the processor (100), cause the processor (100) to receive, via an input processing module (104), user prompts and interaction events from an artificial intelligence agent interface. The processor (100) performs semantic analysis and context extraction on the received user prompts to identify input intent and structure. The processor (100) computes, via a risk computation engine (106), a continuous risk score for each interaction instance using a weighted multi-parameter function incorporating semantic deviation scores, instruction consistency gradients, context sensitivity coefficients, and temporal interaction weights. The processor (100) determines, via a state machine controller (108), a current security state from a plurality of security states based on the computed risk score and temporal interaction patterns, wherein the plurality of security states comprises a safe state, a suspicious state, an under attack state, and a lockdown state. The processor (100) regulates, via a context exposure module (110), access to internal system prompts and memory representations based on the computed risk score and the current security state. The processor (100) executes, via a multi-layer enforcement module (112), a control action based on the current security state, wherein the control action comprises one of allow, sanitize, restrict, or block.

[0020] The technical advantages associated with the above features include the ability to provide continuous risk assessment rather than binary classification, enabling more nuanced security responses to adversarial interactions. The use of a weighted multi-parameter function incorporating multiple input factors allows the system to detect subtle manipulation attempts that may evade single-parameter detection approaches. The state machine-based security management enables persistent tracking of security conditions across multiple interaction instances, thereby addressing staged adversarial attacks that exploit stateless security mechanisms. The multi-layer enforcement architecture provides coordinated protection across different operational layers of the artificial intelligence agent.

[0021] According to other aspects of the present disclosure, the system may include one or more of the following features. The instructions may further cause the processor (100) to maintain an elevated security state despite subsequent low-risk inputs until a predetermined number of consecutive low-risk interactions are received, thereby preventing adversarial bypass through staged interactions.

[0022] The technical advantage associated with maintaining an elevated security state despite subsequent low-risk inputs is that the system resists adversarial strategies that attempt to lower the security posture through intermittent benign inputs. This temporal persistence mechanism prevents attackers from exploiting rapid state transitions to bypass security controls through carefully timed low-risk interactions interspersed with malicious inputs.

[0023] According to other aspects of the present disclosure, the context exposure module (110) may apply selective masking, truncation, or isolation of contextual information when the computed risk score exceeds a predetermined threshold.

[0024] The technical advantage of selective masking, truncation, or isolation of contextual information is that the system can dynamically protect sensitive internal system prompts and memory representations during elevated risk conditions while maintaining normal operational functionality during benign interactions. This approach prevents context leakage attacks that seek to extract

confidential system configuration information through adversarial prompting techniques.

[0025] According to other aspects of the present disclosure, the instructions may further cause the processor (100) to store, via an attack memory module (114),
5 historical adversarial interaction patterns and detected attack signatures, and to update risk computation parameters based on the stored historical adversarial interaction patterns.

[0026] The technical advantage of storing historical adversarial interaction patterns and detected attack signatures is that the system can learn from prior
10 adversarial encounters and improve its detection capabilities over time. By updating risk computation parameters based on stored patterns, the system can proactively identify recurring attack strategies and adapt its security posture to address evolving adversarial techniques.

[0027] According to other aspects of the present disclosure, the multi-layer
15 enforcement module (112) may enforce coordinated security actions across an input filtering layer, a reasoning control layer, a tool execution regulation layer, and an output validation layer under a unified control policy derived from the computed risk score and the current security state.

[0028] The technical advantage of enforcing coordinated security actions
20 across multiple layers under a unified control policy is that the system provides comprehensive protection that addresses adversarial manipulation at different stages of the artificial intelligence agent interaction lifecycle. The unified control policy ensures consistent security enforcement across all layers, preventing attackers from exploiting inconsistencies between independently operating security
25 mechanisms.

[0029] According to other aspects of the present disclosure, the instructions may further cause the processor (100) to track interaction sequences across multiple conversational turns and to detect staged adversarial manipulation patterns based on temporal dependency analysis of the tracked interaction sequences.

30 [0030] The technical advantage of tracking interaction sequences across multiple conversational turns is that the system can detect multi-turn attack

strategies that distribute malicious intent across several seemingly innocuous interactions. Temporal dependency analysis enables identification of staged adversarial manipulation patterns that would not be detectable when analyzing individual interactions in isolation.

5 **[0031]** According to other aspects of the present disclosure, state transitions between the plurality of security states may be triggered when the computed risk score crosses predefined threshold ranges associated with each security state.

10 **[0032]** The technical advantage of threshold-based state transitions is that the system provides predictable and configurable security escalation and de-escalation behavior. The predefined threshold ranges enable administrators to tune the sensitivity of state transitions according to the security requirements of specific deployment environments while maintaining deterministic control over security posture changes.

15 **[0033]** According to other aspects of the present disclosure, the instructions may further cause the processor (100) to monitor system outputs and interaction outcomes via a feedback adaptation mechanism (116) and to adjust the weighted multi-parameter function based on feedback signals derived from the monitored system outputs and interaction outcomes.

20 **[0034]** The technical advantage of monitoring system outputs and adjusting the weighted multi-parameter function based on feedback signals is that the system can continuously refine its risk computation accuracy based on observed outcomes. This feedback-based adaptation enables the system to correct for false positives and false negatives in risk assessment, improving detection performance over time without manual reconfiguration.

25 **[0035]** According to other aspects of the present disclosure, the attack memory module (114) may utilize the stored historical adversarial interaction patterns to enhance risk prediction and enable proactive identification of multi-turn attack strategies.

30 **[0036]** The technical advantage of utilizing stored historical adversarial interaction patterns for enhanced risk prediction is that the system can anticipate and preemptively respond to attack strategies that have been previously observed.

Proactive identification of multi-turn attack strategies enables the system to escalate security measures before an attack sequence reaches completion, thereby preventing successful adversarial manipulation.

5 **[0037]** According to other aspects of the present disclosure, the risk computation engine (106) may aggregate current input signals with historical interaction data to produce a normalized risk value, and the weighted multi-parameter function may be dynamically adjusted using feedback signals derived from prior interaction outcomes.

10 **[0038]** The technical advantage of aggregating current input signals with historical interaction data is that the system can contextualize each interaction within the broader pattern of user behavior, enabling detection of anomalous deviations that may indicate adversarial intent. Dynamic adjustment of the weighted multi-parameter function using feedback signals enables the system to adapt its risk computation to changing threat landscapes and deployment-specific attack patterns without manual intervention.

15 **[0039]** The foregoing general description of the illustrative embodiments and the following detailed description thereof are merely exemplary aspects of the teachings of this disclosure and are not restrictive.

BRIEF DESCRIPTION OF FIGURES

20 **[0040]** The drawings appended herein illustrate exemplary embodiments of the present invention and are provided to enhance understanding of the inventive features disclosed. These figures, when read in conjunction with the detailed description, depict the operational flow, component architecture, and other mechanisms.

25 **[0041]** The illustrations are schematic in nature and serve solely to clarify the principles and functionality of the present disclosure. While particular constructions and configurations are depicted for explanatory purposes, it will be understood that the invention is not limited to the precise implementations shown in the figures and may encompass variations without departing from the scope of protection.

[0042] FIG. 1 is a schematic block diagram of a system for risk-based stateful control of artificial intelligence agent interactions, according to aspects of the present disclosure.

5 [0043] FIG. 2 is a state transition diagram depicting a plurality of security states maintained by a state machine controller, according to an embodiment.

[0044] FIG. 3 is a functional block diagram of a multi-layer enforcement module, according to aspects of the present disclosure.

[0045] FIG. 4 is a schematic diagram of a risk computation engine, according to an embodiment.

10 [0046] FIG. 5 is a flowchart illustrating a method for risk-based stateful control of artificial intelligence agent interactions, according to aspects of the present disclosure.

[0047] FIG. 6 is a schematic diagram of an attack memory module and a feedback adaptation mechanism, according to an embodiment.

15 [0048] Furthermore, in terms of the construction of the system, one or more components of the device may have been represented in the figures by conventional symbols, and the figures may show only those specific details that are pertinent to understanding the embodiments of the present invention so as not to obscure the figures with details that will be readily apparent to those of ordinary skill in the art
20 having benefit of the description herein.

DETAILED DESCRIPTION

[0049] The following detailed description is intended to provide an in-depth understanding of the invention and its various components, configurations, and operational aspects. While specific embodiments are described with reference to
25 the accompanying drawings, the invention is not limited to the particular forms disclosed herein. The scope of the invention shall be interpreted broadly and in accordance with the claims appended hereto.

[0050] The terms “comprises,” “comprising,” “includes,” “including,” and similar expressions used in this description are intended to be open-ended and
30 should be interpreted to include, but not be limited to, the mentioned elements.

Unless explicitly stated otherwise, singular terms may include their plural equivalents and vice versa, depending on the context in which they appear.

[0051] Reference is now made to the accompanying figures, which illustrate example embodiments of the present invention. It should be appreciated that the figures are intended for illustration and explanatory purposes only and do not limit the scope of the invention. Similar reference numerals have been used to denote functionally similar components throughout the figures.

[0052] The embodiments described herein are presented for illustrative purposes and are subject to variations, modifications, and adaptations by those skilled in the art. Any equivalent implementations that perform substantially the same function in substantially the same manner are intended to fall within the scope of this disclosure and the claims appended hereto.

[0053] The use of the term “exemplary” in the description shall be understood to mean “serving as an example or illustration” and shall not be construed as limiting the invention to the preferred embodiments disclosed. Various other embodiments may be developed without departing from the spirit or essential characteristics of the invention. The technical principles and advantages outlined in the foregoing description should be read in conjunction with the figures and claims. It is expressly understood that while the detailed description sets out specific structural and operational aspects, the scope of protection is governed by the claims and includes all legal equivalents of the claimed subject matter. Embodiments of the disclosure are described in the following paragraphs with reference to Figures. In Figures, the same elements or elements that have the same functions are indicated by the same reference signs.

[0054] The present invention will now be described in detail with reference to the accompanying drawings illustrating exemplary embodiments.

[0055] FIG. 1 illustrates a system for risk-based stateful control of artificial intelligence agent interactions according to various embodiments. The system comprises a processor 100 and a memory 102 coupled to processor 100. Memory 102 stores instructions that, when executed by processor 100, cause processor 100 to operate various modules for securing Large Language Model (LLM)-based

artificial intelligence agents against prompt injection, instruction manipulation, and context leakage during interactive operations. Processor 100 may comprise a general-purpose processor, a specialized artificial intelligence accelerator, a multi-core processor, or a distributed processing system, depending on deployment requirements and computational demands.

[0056] The instructions stored in memory 102 cause processor 100 to receive, via an input processing module 104, user prompts and interaction events from an artificial intelligence agent interface. Input processing module 104 performs semantic analysis and context extraction on the received user prompts to identify input intent and structure. Input processing module 104 executes tool invocation checks and validates external action requests as part of input preprocessing. Input processing module 104 logs interaction attributes for risk evaluation and behavioral tracking. Input processing module 104 prepares structured input representations for downstream processing by a risk computation engine 106.

[0057] The instructions further cause processor 100 to compute, via risk computation engine 106, a continuous risk score for each interaction instance using a weighted multi-parameter function. Risk computation engine 106 incorporates semantic deviation scores, instruction consistency gradients, context sensitivity coefficients, and temporal interaction weights in the weighted multi-parameter function. Risk computation engine 106 uses semantic deviation metrics as one of the quantifiable signals for risk evaluation. Risk computation engine 106 uses instruction consistency measures as one of the quantifiable signals for risk evaluation. Risk computation engine 106 uses contextual sensitivity indicators as one of the quantifiable signals for risk evaluation. Risk computation engine 106 uses interaction history representations as one of the quantifiable signals for risk evaluation. Risk computation engine 106 aggregates current input signals with historical interaction data to produce a normalized risk value. The weighted multi-parameter function may be dynamically adjusted using feedback signals derived from prior interaction outcomes.

[0058] The instructions further cause processor 100 to determine, via a state machine controller 108, a current security state from a plurality of security states

based on the computed risk score and temporal interaction patterns. The plurality of security states comprises a safe state, a suspicious state, an under attack state, and a lockdown state. State machine controller 108 governs artificial intelligence agent behavior by dynamically updating the current security state based on

5 computed risk scores, temporal interaction patterns, and feedback signals.

[0059] The instructions further cause processor 100 to regulate, via a context exposure module 110, access to internal system prompts and memory representations based on the computed risk score and the current security state. Context exposure module 110 controls access to user-level data in addition to

10 system-level data based on computed risk and system state. Context exposure module 110 may restrict or permit context exposure to prevent unintended leakage of sensitive system-level or user-level data during adversarial interactions.

[0060] The instructions further cause processor 100 to execute, via a multi-layer enforcement module 112, a control action based on the current security state.

15 The control action comprises one of allow, sanitize, restrict, or block. Multi-layer enforcement module 112 operates under a unified control policy derived from risk and state conditions to ensure consistent and synchronized enforcement across all interaction stages.

[0061] The instructions further cause processor 100 to store, via an attack

20 memory module 114, historical adversarial interaction patterns and detected attack signatures. Attack memory module 114 enables the system to update risk computation parameters based on the stored historical adversarial interaction patterns. Attack memory module 114 supports pattern-based learning for continuous improvement of system defense capabilities.

25 **[0062]** The instructions further cause processor 100 to monitor system outputs and interaction outcomes via a feedback adaptation mechanism 116. Feedback adaptation mechanism 116 adjusts the weighted multi-parameter function based on feedback signals derived from the monitored system outputs and interaction outcomes. Feedback adaptation mechanism 116 incorporates feedback signals to

30 improve decision stability and reduce false positives.

[0063] The system operates within a unified event-driven control pipeline enabling continuous adaptive and state-aware enforcement of artificial intelligence agent security suitable for real-time deployment in interactive and autonomous system environments. The system regulates artificial intelligence agent behavior as
5 a continuous control process rather than a post-processing or filtering task. The system implements a closed-loop control architecture wherein computed risk values directly influence system state transitions and context access decisions in real time. The integration of risk computation engine 106 with state machine controller 108 and context exposure module 110 forms a tightly coupled control loop that enables
10 real-time adaptive enforcement across multiple system layers.

[0064] FIG. 2 illustrates a state transition diagram depicting a plurality of security states maintained by state machine controller 108 according to various embodiments. The plurality of security states comprises a safe state 202, a suspicious state 204, an under attack state 206, and a lockdown state 208. State
15 machine controller 108 manages transitions between safe state 202, suspicious state 204, under attack state 206, and lockdown state 208 based on computed risk scores and temporal interaction characteristics.

[0065] State transitions between the plurality of security states are triggered when the computed risk score crosses predefined threshold ranges associated with
20 each security state. Safe state 202 represents a baseline operational condition in which the artificial intelligence agent operates with standard permissions and context access. When the computed risk score exceeds a first threshold, state machine controller 108 transitions from safe state 202 to suspicious state 204. Suspicious state 204 represents an elevated monitoring condition in which state
25 machine controller 108 applies increased scrutiny to incoming interactions. When the computed risk score exceeds a second threshold, state machine controller 108 transitions from suspicious state 204 to under attack state 206. Under attack state 206 represents a condition in which state machine controller 108 applies restrictive controls to mitigate detected adversarial activity. When the computed risk score
30 exceeds a third threshold, state machine controller 108 transitions from under attack state 206 to lockdown state 208. Lockdown state 208 represents a condition in

which state machine controller 108 blocks or severely restricts artificial intelligence agent operations to prevent further adversarial manipulation.

[0066] State machine controller 108 performs state transitions based on threshold-based rules derived from risk score ranges. The threshold-based rules
5 define numerical boundaries that separate each security state from adjacent security states. In some cases, the thresholds may comprise fixed thresholds that remain constant throughout system operation. In some cases, the thresholds may comprise adaptive thresholds that are adjusted based on historical interaction patterns stored in attack memory module 114. In some cases, the thresholds may comprise
10 dynamically computed thresholds that are calculated based on environmental conditions such as system load, time of day, or deployment context.

[0067] State machine controller 108 performs state transitions based on condition-driven rules derived from temporal interaction characteristics. The condition-driven rules evaluate patterns across multiple interaction instances to
15 detect staged adversarial manipulation. State machine controller 108 may analyze the rate of risk score changes, the frequency of elevated risk interactions, and the distribution of risk scores over a sliding time window to inform state transition decisions.

[0068] State machine controller 108 implements temporal persistence such that
20 the system maintains an elevated security state based on prior high-risk interactions even when subsequent inputs exhibit low risk. The instructions cause processor 100 to maintain an elevated security state despite subsequent low-risk inputs until a predetermined number of consecutive low-risk interactions are received, thereby preventing adversarial bypass through staged interactions. Temporal persistence
25 prevents immediate state reversal when an adversary attempts to lower the security posture by interspersing benign inputs between malicious inputs. State machine controller 108 may require a configurable number of consecutive low-risk interactions before permitting de-escalation from an elevated security state to a lower security state.

30 [0069] The system implements controlled recovery from lockdown state 208 through gradual safe inputs rather than immediate state reversal. When the system

enters lockdown state 208, state machine controller 108 may require an extended sequence of verified low-risk interactions before transitioning to under attack state 206. State machine controller 108 may further require additional consecutive low-risk interactions before transitioning from under attack state 206 to suspicious state 204, and from suspicious state 204 to safe state 202. The controlled recovery mechanism prevents adversaries from rapidly cycling the system through security states to exploit transient vulnerabilities during state transitions.

[0070] In some cases, state machine controller 108 may implement asymmetric transition requirements wherein escalation to higher security states occurs more rapidly than de-escalation to lower security states. In some cases, state machine controller 108 may apply different threshold configurations for different categories of artificial intelligence agent operations or different user privilege levels. In some cases, state machine controller 108 may receive threshold adjustment signals from feedback adaptation mechanism 116 to refine state transition behavior based on observed system performance and false positive rates.

[0071] FIG. 3 illustrates a functional block diagram of multi-layer enforcement module 112 according to various embodiments. Multi-layer enforcement module 112 comprises an input filtering layer 302, a reasoning control layer 304, a tool execution regulation layer 306, and an output validation layer 308 operating in coordinated sequence. Multi-layer enforcement module 112 further comprises a unified control policy 310 that receives the computed risk score from risk computation engine 106 and the current security state from state machine controller 108. Unified control policy 310 derives a unified control policy governing input filtering layer 302, reasoning control layer 304, tool execution regulation layer 306, and output validation layer 308 such that coordinated security actions are enforced across all four layers.

[0072] Multi-layer enforcement module 112 enforces coordinated security actions across input filtering layer 302, reasoning control layer 304, tool execution regulation layer 306, and output validation layer 308 under unified control policy 310 derived from the computed risk score and the current security state. Multi-layer enforcement module 112 operates under interdependent enforcement with shared

control signals between layers. The shared control signals enable each layer to receive state information and enforcement directives from unified control policy 310, and to propagate enforcement outcomes to subsequent layers in the coordinated sequence.

5 **[0073]** Input filtering layer 302 receives user prompts and interaction events and performs input sanitization as a coordinated security action across system layers. Input filtering layer 302 may employ pattern matching to identify and filter known adversarial input patterns. In some cases, input filtering layer 302 may employ machine learning classifiers to detect anomalous input characteristics. In
10 some cases, input filtering layer 302 may employ rule-based filters to enforce input format and content constraints. Input filtering layer 302 applies sanitization operations based on control signals received from unified control policy 310, wherein the sanitization operations may include character escaping, token removal, or input restructuring depending on the current security state.

15 **[0074]** Reasoning control layer 304 receives processed inputs from input filtering layer 302 and enforces reasoning control and constraint enforcement as a coordinated security action. Reasoning control layer 304 governs the reasoning processes of the artificial intelligence agent by applying constraints that limit reasoning scope, depth, or direction based on the current security state. In some
20 cases, reasoning control layer 304 may use constraint solvers to enforce logical boundaries on reasoning operations. In some cases, reasoning control layer 304 may use logic gates to selectively enable or disable reasoning pathways. In some cases, reasoning control layer 304 may use attention masking to restrict the artificial intelligence agent from attending to specific portions of the input or context during
25 reasoning operations.

[0075] Tool execution regulation layer 306 receives reasoning outputs from reasoning control layer 304 and enforces tool execution regulation as a coordinated security action. Tool execution regulation layer 306 governs access to external tools, application programming interfaces, and system resources that the artificial
30 intelligence agent may invoke during interaction processing. In some cases, tool execution regulation layer 306 may implement whitelisting to permit access to a

predefined set of approved tools based on the current security state. In some cases, tool execution regulation layer 306 may implement sandboxing to isolate tool execution within restricted environments that limit potential damage from adversarial tool invocations. In some cases, tool execution regulation layer 306 may
5 implement permission-based access control to grant or deny tool access based on user privilege levels and computed risk scores.

[0076] Output validation layer 308 receives outputs from tool execution regulation layer 306 and enforces output validation and filtering as a coordinated security action. Output validation layer 308 examines artificial intelligence agent
10 responses before delivery to users or downstream systems. In some cases, output validation layer 308 may use content filters to detect and remove sensitive information, adversarial content, or policy-violating material from outputs. In some cases, output validation layer 308 may use format validators to ensure outputs conform to expected structural and syntactic requirements. In some cases, output
15 validation layer 308 may use semantic analyzers to evaluate output meaning and intent for consistency with system policies and user expectations.

[0077] Unified control policy 310 coordinates the enforcement actions of input filtering layer 302, reasoning control layer 304, tool execution regulation layer 306, and output validation layer 308 based on the computed risk score and the current
20 security state. Unified control policy 310 generates control signals that specify enforcement intensity, permitted operations, and constraint parameters for each layer. When the current security state corresponds to safe state 202, unified control policy 310 may configure each layer to operate with standard enforcement parameters. When the current security state corresponds to suspicious state 204,
25 unified control policy 310 may configure each layer to apply elevated scrutiny and additional validation checks. When the current security state corresponds to under attack state 206, unified control policy 310 may configure each layer to apply restrictive enforcement that limits artificial intelligence agent capabilities. When the current security state corresponds to lockdown state 208, unified control policy
30 310 may configure each layer to block or severely restrict operations.

[0078] The interdependent enforcement architecture enables each layer to communicate enforcement outcomes and detected anomalies to unified control policy 310. In some cases, input filtering layer 302 may signal detection of suspicious input patterns to unified control policy 310, which may in turn adjust control signals to reasoning control layer 304, tool execution regulation layer 306, and output validation layer 308. In some cases, output validation layer 308 may signal detection of policy-violating outputs to unified control policy 310, which may trigger feedback to risk computation engine 106 for risk score adjustment. The shared control signals between layers enable coordinated escalation and de-escalation of enforcement actions across the multi-layer enforcement module 112.

[0079] FIG. 4 illustrates a schematic diagram of risk computation engine 106 according to various embodiments. Risk computation engine 106 receives a semantic deviation score input 402, an instruction consistency gradient input 404, a context sensitivity coefficient input 406, and a temporal interaction weight input 408 as inputs to a weighted multi-parameter aggregator 410. Semantic deviation score input 402 represents a quantifiable signal derived from semantic deviation metrics that measure divergence between received user prompts and expected input patterns. Instruction consistency gradient input 404 represents a quantifiable signal derived from instruction consistency measures that evaluate coherence between user instructions and established system directives. Context sensitivity coefficient input 406 represents a quantifiable signal derived from contextual sensitivity indicators that assess the degree to which an interaction attempts to access or manipulate sensitive system context. Temporal interaction weight input 408 represents a quantifiable signal derived from interaction history representations that capture patterns across multiple interaction instances over time.

[0080] Weighted multi-parameter aggregator 410 aggregates semantic deviation score input 402, instruction consistency gradient input 404, context sensitivity coefficient input 406, and temporal interaction weight input 408 using a weighted multi-parameter function. Weighted multi-parameter aggregator 410 outputs a normalized risk value 412 representing the continuous risk score for each interaction instance. Normalized risk value 412 provides a numerical representation

of assessed risk that state machine controller 108 uses to determine the current security state and that multi-layer enforcement module 112 uses to select appropriate control actions.

[0081] Risk computation engine 106 derives the risk score using normalized feature vectors incorporating semantic deviation scores, instruction consistency gradients, context sensitivity coefficients, and temporal interaction weights. In some cases, risk computation engine 106 may compute the risk score using a learned model in addition to or instead of a weighted scoring function to produce normalized risk value 412 for each interaction instance. The learned model may comprise a machine learning model trained on labeled adversarial and benign interaction data to distinguish between safe and malicious inputs.

[0082] Weighted multi-parameter aggregator 410 aggregates current input signals with historical interaction data to produce normalized risk value 412. The aggregation of current input signals with historical interaction data enables risk computation engine 106 to contextualize each interaction within the broader pattern of user behavior across multiple conversational turns.

[0083] In some cases, weighted multi-parameter aggregator 410 may implement a linear weighted sum that combines semantic deviation score input 402, instruction consistency gradient input 404, context sensitivity coefficient input 406, and temporal interaction weight input 408 using configurable weight coefficients. In some cases, weighted multi-parameter aggregator 410 may implement neural network-based aggregation that processes the inputs through one or more neural network layers to produce normalized risk value 412. In some cases, weighted multi-parameter aggregator 410 may implement ensemble methods that combine outputs from multiple aggregation approaches to improve risk assessment accuracy. In some cases, weighted multi-parameter aggregator 410 may implement fuzzy logic aggregation that applies fuzzy membership functions and fuzzy rules to handle uncertainty in input signals.

[0084] Risk computation engine 106 computes tool invocation risk as an elevated indicator for identifying unsafe tool usage. When an interaction includes requests to invoke external tools, application programming interfaces, or system

resources, risk computation engine 106 may assign elevated risk scores to such interactions based on the nature of the requested tool, the context of the request, and the current security state. Tool invocation risk enables the system to detect adversarial attempts to exploit tool access for unauthorized operations.

5 **[0085]** Feedback adaptation mechanism 116 supplies a feedback adjustment path 414 to weighted multi-parameter aggregator 410. Feedback adjustment path 414 enables the weighted multi-parameter function to be dynamically adjusted based on feedback signals derived from monitored system outputs and prior interaction outcomes. Feedback adaptation mechanism 116 receives monitored
10 outputs and interaction outcomes from risk computation engine 106 and generates feedback signals that modify the behavior of weighted multi-parameter aggregator 410.

[0086] Feedback adaptation mechanism 116 adjusts state transition thresholds based on observed system behavior to improve decision stability. When feedback
15 adaptation mechanism 116 detects that state machine controller 108 exhibits excessive state oscillation or unstable transition behavior, feedback adaptation mechanism 116 may adjust the threshold values that govern state transitions to reduce sensitivity to transient risk fluctuations.

[0087] Feedback adaptation mechanism 116 incorporates feedback signals to
20 reduce false positives in risk detection. When feedback adaptation mechanism 116 identifies that risk computation engine 106 produces elevated risk scores for interactions that are subsequently determined to be benign, feedback adaptation mechanism 116 may adjust the weight coefficients or model parameters of weighted multi-parameter aggregator 410 to reduce the occurrence of false positive
25 risk assessments.

[0088] The system performs parameter tuning for risk scoring weights and state thresholds as part of optimization. Parameter tuning may comprise adjustment of the weight coefficients applied to semantic deviation score input 402, instruction consistency gradient input 404, context sensitivity coefficient input 406, and
30 temporal interaction weight input 408. Parameter tuning may further comprise

adjustment of the threshold values that define boundaries between safe state 202, suspicious state 204, under attack state 206, and lockdown state 208.

[0089] The system monitors system consistency and response accuracy during operation. Monitoring of system consistency enables feedback adaptation mechanism 116 to detect drift in risk computation behavior over time. Monitoring of response accuracy enables feedback adaptation mechanism 116 to evaluate whether control actions selected by multi-layer enforcement module 112 produce intended security outcomes. The monitored consistency and accuracy metrics inform the feedback signals that feedback adaptation mechanism 116 supplies via feedback adjustment path 414 to weighted multi-parameter aggregator 410.

[0090] FIG. 5 illustrates a flowchart of a method 500 performed by the system for risk-based stateful control of artificial intelligence agent interactions according to various embodiments. Method 500 comprises a step of receiving user prompts and interaction events 502, a step of performing semantic analysis and context extraction 504, a step of computing continuous risk score 506, a step of determining current security state 508, a step of regulating context exposure 510, a step of executing control action 512, and a step of updating attack memory and feedback parameters 514.

[0091] At step of receiving user prompts and interaction events 502, input processing module 104 receives user prompts and interaction events from an artificial intelligence agent interface. The received user prompts may include text inputs, voice inputs converted to text, or structured command inputs. The interaction events may include tool invocation requests, context access requests, or session management events.

[0092] At step of performing semantic analysis and context extraction 504, input processing module 104 performs semantic analysis and context extraction on the received user prompts to identify input intent and structure. Semantic analysis may comprise parsing the received user prompts to extract semantic features, intent indicators, and structural characteristics. Context extraction may comprise identifying references to prior interactions, system state, or external resources within the received user prompts.

[0093] At step of computing continuous risk score 506, risk computation engine 106 computes a continuous risk score for each interaction instance using a weighted multi-parameter function. The weighted multi-parameter function incorporates semantic deviation scores, instruction consistency gradients, context sensitivity coefficients, and temporal interaction weights. Risk computation engine 106 aggregates current input signals with historical interaction data to produce a normalized risk value representing the continuous risk score.

[0094] At step of determining current security state 508, state machine controller 108 determines a current security state from a plurality of security states based on the computed risk score and temporal interaction patterns. The plurality of security states comprises a safe state, a suspicious state, an under attack state, and a lockdown state. State machine controller 108 evaluates the computed risk score against predefined threshold ranges to determine whether state transitions are triggered.

[0095] At step of regulating context exposure 510, context exposure module 110 regulates access to internal system prompts and memory representations based on the computed risk score and the current security state. Context exposure module 110 applies selective masking of contextual information under elevated risk conditions to prevent unintended information disclosure. Context exposure module 110 applies truncation of contextual information under elevated risk conditions to prevent unintended information disclosure. Context exposure module 110 applies isolation of contextual information under elevated risk conditions to prevent unintended information disclosure. When the computed risk score exceeds a predetermined threshold, context exposure module 110 may apply selective masking, truncation, or isolation of contextual information to protect sensitive system data from adversarial extraction attempts.

[0096] At step of executing control action 512, multi-layer enforcement module 112 executes a control action based on the current security state. The control action comprises one of allow, sanitize, restrict, or block. Multi-layer enforcement module 112 selects the control action based on the current security state and the computed risk score, and coordinates enforcement across input

filtering layer 302, reasoning control layer 304, tool execution regulation layer 306, and output validation layer 308.

[0097] At step of updating attack memory and feedback parameters 514, attack memory module 114 stores detected adversarial interaction patterns and attack signatures, and feedback adaptation mechanism 116 adjusts risk computation parameters based on monitored outcomes. The updating of attack memory and feedback parameters enables the system to learn from prior adversarial encounters and improve detection capabilities over time.

[0098] FIG. 6 illustrates a schematic diagram of attack memory module 114 and feedback adaptation mechanism 116 according to various embodiments. Attack memory module 114 comprises a historical adversarial pattern store 602, an attack signature store 604, and a multi-turn sequence tracker 606. Feedback adaptation mechanism 116 receives stored data from attack memory module 114 and supplies a parameter update path 608 to risk computation engine 106.

[0099] Historical adversarial pattern store 602 stores historical adversarial interaction patterns detected during prior system operations. The stored historical adversarial interaction patterns may include input sequences, behavioral characteristics, and contextual features associated with previously identified adversarial interactions. Historical adversarial pattern store 602 enables the system to compare current interactions against known adversarial patterns for pattern matching and anomaly detection.

[0100] Attack signature store 604 stores detected attack signatures representing characteristic features of identified attack types. In some cases, attack signature store 604 may store hash-based signatures that represent compact fingerprints of known attack patterns. In some cases, attack signature store 604 may store vector embeddings that represent attack patterns in a continuous vector space for similarity-based matching. In some cases, attack signature store 604 may store pattern templates that define structural characteristics of attack sequences. In some cases, attack signature store 604 may store behavioral fingerprints that capture temporal and behavioral characteristics of adversarial interactions.

[0101] Multi-turn sequence tracker 606 tracks interaction sequences across multiple conversational turns to detect staged adversarial manipulation patterns based on temporal dependency analysis of the tracked interaction sequences. Multi-turn sequence tracker 606 identifies staged and evolving adversarial manipulation patterns that cannot be identified through single-step analysis. In some cases, multi-turn sequence tracker 606 may implement sequence models that process interaction sequences to detect temporal dependencies indicative of staged attacks. In some cases, multi-turn sequence tracker 606 may implement temporal convolutional networks that apply convolutional operations across time to identify attack patterns spanning multiple turns. In some cases, multi-turn sequence tracker 606 may implement attention-based trackers that apply attention mechanisms to weight the relevance of prior interactions when evaluating current interaction risk.

[0102] Parameter update path 608 enables risk computation parameters to be updated based on stored historical adversarial interaction patterns. Feedback adaptation mechanism 116 analyzes the stored data from historical adversarial pattern store 602, attack signature store 604, and multi-turn sequence tracker 606 to generate parameter updates that improve risk computation accuracy. Parameter update path 608 supplies the generated parameter updates to risk computation engine 106, whereby the weighted multi-parameter function is adjusted based on learned adversarial patterns.

[0103] Attack memory module 114 supports pattern-based learning for continuous improvement of system defense capabilities. Pattern-based learning enables the system to refine detection models and risk computation parameters based on accumulated adversarial interaction data. Attack memory module 114 enables proactive identification of staged attack strategies through stored adversarial patterns. Attack memory module 114 utilizes the stored historical adversarial interaction patterns to enhance risk prediction and enable proactive identification of multi-turn attack strategies. The proactive identification capability enables the system to escalate security measures before an attack sequence reaches completion.

[0104] The system uses a training validation testing data partitioning ratio of 60/20/20 while maintaining temporal ordering to preserve realistic interaction sequences. The 60/20/20 partitioning allocates 60 percent of interaction data for training risk computation models, 20 percent for validation during model development, and 20 percent for testing model performance. Maintaining temporal ordering ensures that training data precedes validation data and validation data precedes testing data, thereby preventing information leakage from future interactions into model training and preserving realistic evaluation conditions.

[0105] It is to be appreciated that the concepts, systems, and techniques sought to be protected herein are not limited to use in the example applications described herein, but rather may be useful in substantially any application where it is desired to secure artificial intelligence agent interactions against adversarial manipulation. The system for risk-based stateful control of artificial intelligence agent interactions is susceptible of industrial application in a wide range of industries and deployment environments. In some aspects, the system may be deployed in customer service and support industries where LLM-based conversational agents interact with users and handle sensitive customer information. In some aspects, the system may be deployed in financial services industries where artificial intelligence agents process transactions, provide financial advice, or access confidential account information. In some aspects, the system may be deployed in healthcare industries where artificial intelligence agents interact with patients, access medical records, or provide health-related information. In some aspects, the system may be deployed in enterprise software environments where artificial intelligence agents have access to internal corporate systems, databases, and proprietary information. In some aspects, the system may be deployed in autonomous agent platforms where artificial intelligence agents execute tool invocations and interact with external application programming interfaces. In some aspects, the system may be deployed in educational technology platforms where artificial intelligence agents interact with students and access educational content. In some aspects, the system may be deployed in government and public sector applications where artificial intelligence agents handle citizen inquiries and access sensitive governmental data. While

particular embodiments and applications of the present disclosure have been illustrated and described in connection with security for LLM-based artificial intelligence agents, it is to be understood that embodiments of the disclosure are not limited to the precise constructions and compositions disclosed herein and that
5 various modifications, changes, and variations can be apparent from the foregoing descriptions without departing from the spirit and scope of the disclosure as defined in the appended claims.

[0106] There are several technical advantages of the system for risk-based stateful control of artificial intelligence agent interactions of the present
10 disclosure. First, the use of continuous risk scoring rather than binary classification enables more nuanced security responses to adversarial interactions, allowing the system to detect subtle manipulation attempts that may evade single-parameter detection approaches. Second, the state machine-based security management enables persistent tracking of security conditions across multiple interaction
15 instances, thereby addressing staged adversarial attacks that exploit stateless security mechanisms. Third, the multi-layer enforcement architecture provides coordinated protection across input filtering, reasoning control, tool execution regulation, and output validation layers under a unified control policy, preventing attackers from exploiting inconsistencies between independently operating security
20 mechanisms. Fourth, the temporal persistence mechanism maintains an elevated security state despite subsequent low-risk inputs until a predetermined number of consecutive low-risk interactions are received, thereby resisting adversarial strategies that attempt to lower the security posture through intermittent benign inputs. Fifth, the context exposure regulation mechanism dynamically protects
25 sensitive internal system prompts and memory representations during elevated risk conditions while maintaining normal operational functionality during benign interactions, preventing context leakage attacks that seek to extract confidential system configuration information. Sixth, the attack memory module stores historical adversarial interaction patterns and detected attack signatures, enabling
30 the system to learn from prior adversarial encounters and proactively identify recurring attack strategies. Seventh, the feedback adaptation mechanism enables the

system to continuously refine its risk computation accuracy based on observed outcomes, correcting for false positives and false negatives in risk assessment without manual reconfiguration. These technical advantages collectively contribute to a comprehensive solution for securing artificial intelligence agent interactions
5 that is both adaptive and robust against evolving adversarial techniques.

[0107] It will be appreciated that the above-detailed description, along with FIGs, is provided to illustrate the architecture and operation of exemplary embodiments of the present invention. Various modifications and enhancements can be made without departing from the scope of the invention. All such variations,
10 as would be recognized by those skilled in the art, are intended to be within the scope of the present disclosure, which is defined by the claims that follow.

[0108] It will be appreciated by persons skilled in the art that while the invention has been described with reference to specific embodiments and accompanying drawings, numerous modifications, substitutions, variations, and
15 equivalents are possible without departing from the spirit or scope of the present invention. The use of singular terms such as “a,” “an,” and “the” is not intended to limit the disclosed elements to a single instance unless explicitly stated, and such terms should be interpreted to include plural forms as applicable. Likewise, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having,” and
20 their grammatical variants are intended to be open-ended and non-limiting, and should be interpreted as meaning “including but not limited to.” Any enumerated listing of components, features, or elements does not imply that the items are mutually exclusive unless expressly stated.

[0109] It is further understood that where features, characteristics, or elements
25 are described in connection with a Markush group or a disjunctive phrase (e.g., “A or B”), the invention includes all possible combinations and sub-combinations thereof unless explicitly excluded. Thus, a statement referring to “at least one of A, B, and C” should be interpreted to mean any one or more of A, B, and C, individually or in any combination. The language used in this specification is
30 selected primarily for clarity and illustrative purposes and should not be construed to limit the inventive scope unless specifically recited in the claims. Accordingly,

the foregoing description of embodiments should be regarded as illustrative and not restrictive, with the scope of the invention being defined solely by the appended claims and their legal equivalents

[0110] It will be recognized that various features, elements, and combinations thereof described herein may be desirably adapted for alternative implementations or other applications. Many such modifications, substitutions, enhancements, or equivalents may become apparent to those skilled in the art upon reading this disclosure and are considered to fall within the scope and spirit of the present invention. The invention is not limited to the precise configurations or exemplary embodiments set forth in this specification, and various alternatives may be used without departing from the intended objectives. The claims, and not the detailed description, shall define the legal scope of protection afforded by the present disclosure.

[0111] In interpreting the specification and claims, all terms should be construed in the broadest reasonable manner consistent with the context and the understanding of those skilled in the art. The use of terms such as “comprises,” “comprising,” “includes,” “including,” or “has” should be interpreted to be non-exclusive and not limited to the stated elements alone. Moreover, references to “one embodiment,” “an embodiment,” or “in certain embodiments” are not meant to imply that such features are required or exclusive, and any feature described in connection with one embodiment may be used in combination with other features or embodiments unless clearly prohibited or contextually inconsistent.

[0112] It should be further appreciated that reference throughout the specification to features, operations, or components using singular terms should not be construed as excluding a plurality thereof unless the context expressly indicates otherwise. Similarly, features described in the context of grouped or listed elements (e.g., A, B, and C) should be understood to encompass individual, multiple, or all combinations of said elements. Where ranges are given, all intermediate values and subranges are understood to be disclosed as if specifically recited. The scope of the invention should therefore be construed as inclusive of all such combinations and

logical extensions as would be appreciated by a person of ordinary skill in the relevant technical field

[0113] In another embodiment, the above disclosure is a description of the invention and is not intended to limit the scope of the invention. Other variations and modifications of the above-described embodiment shall be apparent to those skilled in the art and are intended to fall within the scope of the invention as defined in the following claims.

Dated This day of May 2026



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CLAIMS

WE CLAIM:

1. A system for risk-based stateful control of artificial intelligence agent interactions, comprising:
 - 5 a processor (100); and
 - a memory (102) coupled to the processor (100), the memory (102) storing instructions that, when executed by the processor (100), cause the processor (100) to:
 - 10 receive, via an input processing module (104), user prompts and interaction events from an artificial intelligence agent interface;
 - perform semantic analysis and context extraction on the received user prompts to identify input intent and structure;
 - 15 compute, via a risk computation engine (106), a continuous risk score for each interaction instance using a weighted multi-parameter function incorporating semantic deviation scores, instruction consistency gradients, context sensitivity coefficients, and temporal interaction weights;
 - determine, via a state machine controller (108), a current security state from a plurality of security states based on the computed risk score and temporal interaction patterns, wherein the plurality of security states
20 comprises a safe state, a suspicious state, an under attack state, and a lockdown state;
 - regulate, via a context exposure module (110), access to internal system prompts and memory representations based on the computed risk score and the current security state; and
 - 25 execute, via a multi-layer enforcement module (112), a control action based on the current security state, wherein the control action comprises one of allow, sanitize, restrict, or block.
2. The system of claim 1, wherein the instructions further cause the
30 processor (100) to maintain an elevated security state despite subsequent low-risk inputs until a predetermined number of consecutive low-risk interactions are received, thereby preventing adversarial bypass through staged interactions.

3. The system of claim 1, wherein the context exposure module (110) applies selective masking, truncation, or isolation of contextual information when the computed risk score exceeds a predetermined threshold.

4. The system of claim 1, wherein the instructions further cause the processor (100) to:

store, via an attack memory module (114), historical adversarial interaction patterns and detected attack signatures; and

update risk computation parameters based on the stored historical adversarial interaction patterns.

5. The system of claim 1, wherein the multi-layer enforcement module (112) enforces coordinated security actions across an input filtering layer, a reasoning control layer, a tool execution regulation layer, and an output validation layer under a unified control policy derived from the computed risk score and the current security state.

6. The system of claim 1, wherein the instructions further cause the processor (100) to:

track interaction sequences across multiple conversational turns; and

detect staged adversarial manipulation patterns based on temporal dependency analysis of the tracked interaction sequences.

7. The system of claim 1, wherein state transitions between the plurality of security states are triggered when the computed risk score crosses predefined threshold ranges associated with each security state.

8. The system of claim 1, wherein the instructions further cause the processor (100) to:

monitor system outputs and interaction outcomes via a feedback adaptation mechanism (116); and

adjust the weighted multi-parameter function based on feedback signals derived from the monitored system outputs and interaction outcomes.

9. The system of claim 4, wherein the attack memory module (114) utilizes the stored historical adversarial interaction patterns to enhance risk prediction and enable proactive identification of multi-turn attack strategies.

10. The system of claim 1, wherein the risk computation engine (106) aggregates current input signals with historical interaction data to produce a normalized risk value, and wherein the weighted multi-parameter function is dynamically adjusted using feedback signals derived from prior interaction outcomes.

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ABSTRACT

SYSTEM FOR RISK-BASED STATEFUL CONTROL OF ARTIFICIAL INTELLIGENCE AGENT INTERACTIONS

A system for risk-based stateful control of artificial intelligence agent interactions
5 comprises a processor (100) and memory (102) storing instructions causing the
processor (100) to receive user prompts and interaction events via an input
processing module (104), perform semantic analysis and context extraction to
identify input intent and structure, compute via a risk computation engine (106) a
continuous risk score using a weighted multi-parameter function incorporating
10 semantic deviation scores, instruction consistency gradients, context sensitivity
coefficients, and temporal interaction weights, determine via a state machine
controller (108) a current security state from safe, suspicious, under attack, and
lockdown states based on the computed risk score and temporal interaction patterns,
regulate via a context exposure module (110) access to internal system prompts and
15 memory representations, and execute via a multi-layer enforcement module (112)
a control action comprising allow, sanitize, restrict, or block.

(Fig. 1)

Dated This day of May 2026



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